

Technological Framework for Forecasting Cryptocurrency Prices using the Long Short-Term Memory Algorithm

by David Riley



- Motivation
 - Machine Learning Experience
 - Automated Trading System
- Requirements
 - Collection of real time market data
 - Persistent storage of market data
 - Data Aggregation
 - Model Development and Feature Engineering
 - Model Training and Experiment Tracking
 - Model Execution
 - Backend
 - Frontend

Time Series Forecasting:

- Autoformer (1)
- Long Short-Term Memory (LSTM)
 - Simplicity (High Level Libraries - TensorFlow/Keras)
 - Resources / Community Support
 - Adaptable to varying feature sets

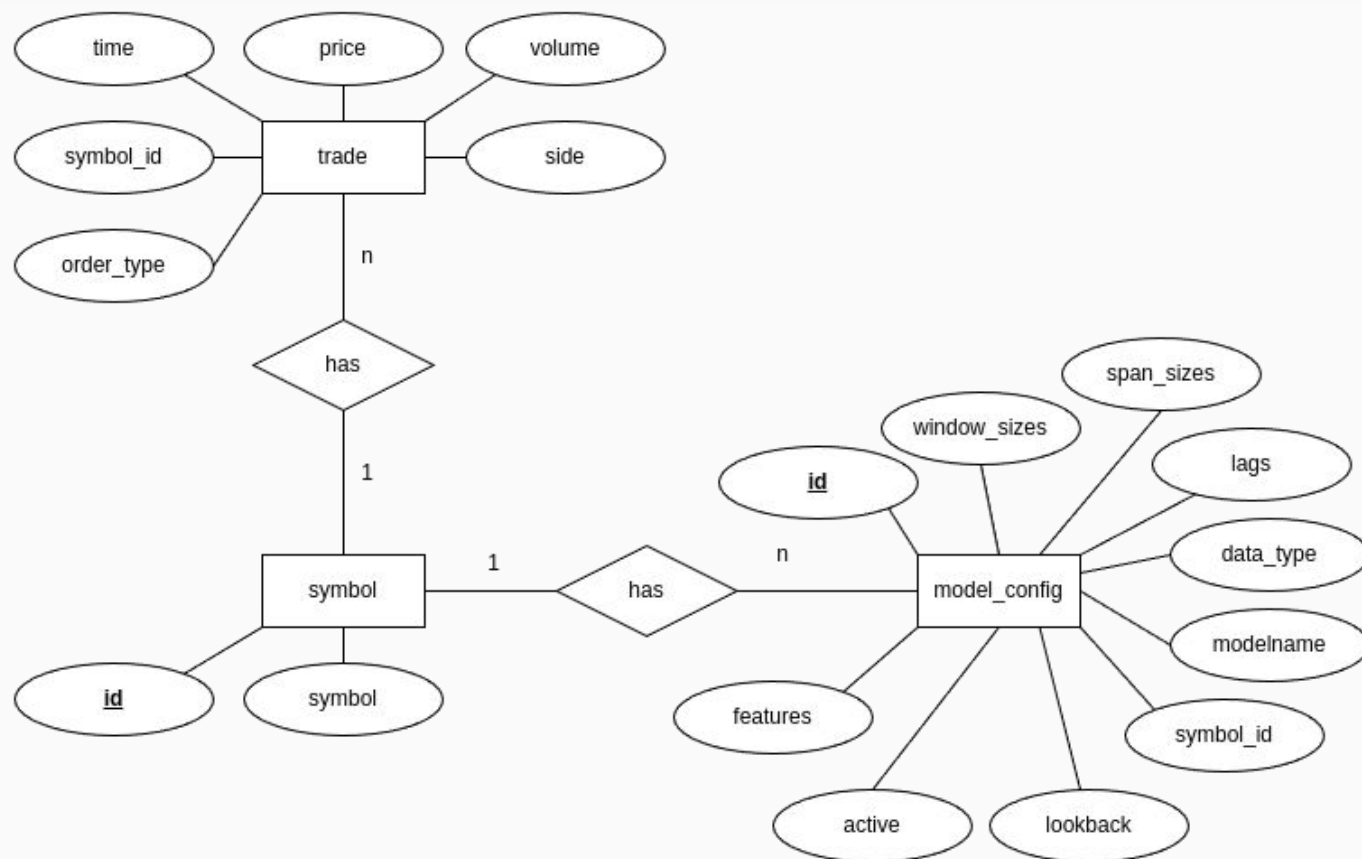
Infrastructure:

- Databricks & MIFlow
- PostgresML
- Micro-Service Architecture

Technology Stack

- Timescale/PostgreSQL
- MIFlow - Model Training and Experiment Tracking
- MinIO - S3 Object Store
- Model Service (Python)
- Backend (Rust)
- Frontend (Angular 17 / Node.js)

ER-Diagram



Design and Implementation: Data Flow

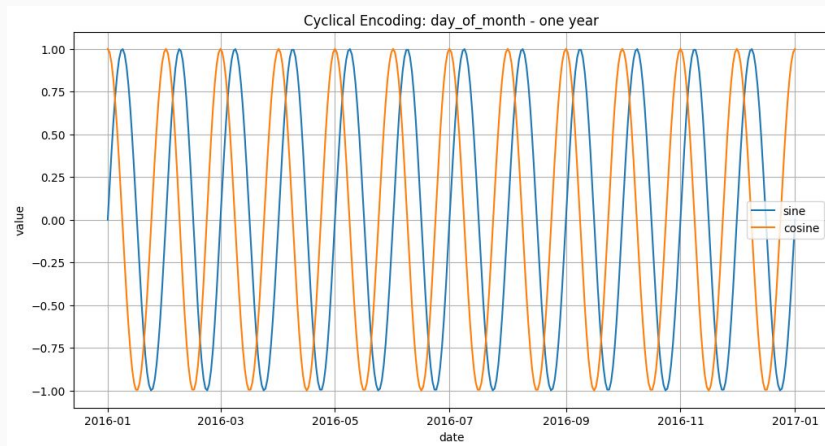
- Collector (Kraken Websocket Api)
- Trade Table (Hypertable)
- Materialized View
 - Hourly Time Buckets
 - Open-, high-, low-, close-price, volume, trade count
- Model Service
 - Executes LSTM Model
- Backend
- Frontend Dashboard
 - Visualization of past values alongside with the forecasted close-price

```
1 CREATE MATERIALIZED VIEW ohlc_hour
2 WITH (timescaledb.continuous, timescaledb.create_group_indexes=TRUE,
   timescaledb.finalized=TRUE) AS
3 SELECT
4     time_bucket('1 hour', time) AS bucket,
5     symbol_id,
6     FIRST(price, time) AS open_price,
7     MAX(price) AS high,
8     MIN(price) AS low,
9     LAST(price, time) AS close_price,
10    SUM(volume) AS volume,
11    CAST(COUNT(*) AS INTEGER) AS count
12 FROM trade
13 GROUP BY bucket, symbol_id;
```

Design and Implementation: LSTM

- Feature Engineering

- Lags, Simple/Exponential Moving Averages, Holidays, Volume, Trade Count
- Cyclic Encoding (day of the month, hours, etc.)



- Model Tuning Script (Hyperparameter Tuning)

- Parameter Grid: learning rate, optimizer, epochs, loss, lookback, ...
- Tracking using MIFlow: Metrics, Plots, Model Architecture, Model

MLflow: Dashboard

MLflow

localhost:5000/#/experiments/0?searchFilter=&orderByKey=metrics.%60mape%60&orderByAsc=true&startTime=ALL&lifecycleFilter=Active&datasetsFilter=W10%3D8&modelVersionFilter=All%20Runs&sel...

New Chrome available

mlflow 2.10.2

Experiments Models

Toggle Dark Mode GitHub Docs

Experiments

Search Experiments

Default

Default

Provide Feedback

Experiment ID: 0

Artifact Location: s3://mlflow-models/0

Description Edit

metrics.rmse < 1 and params.model = "tree"

Time created

State: Active

Datasets

Sort: mape

Columns

New run

Table

Chart

Evaluation

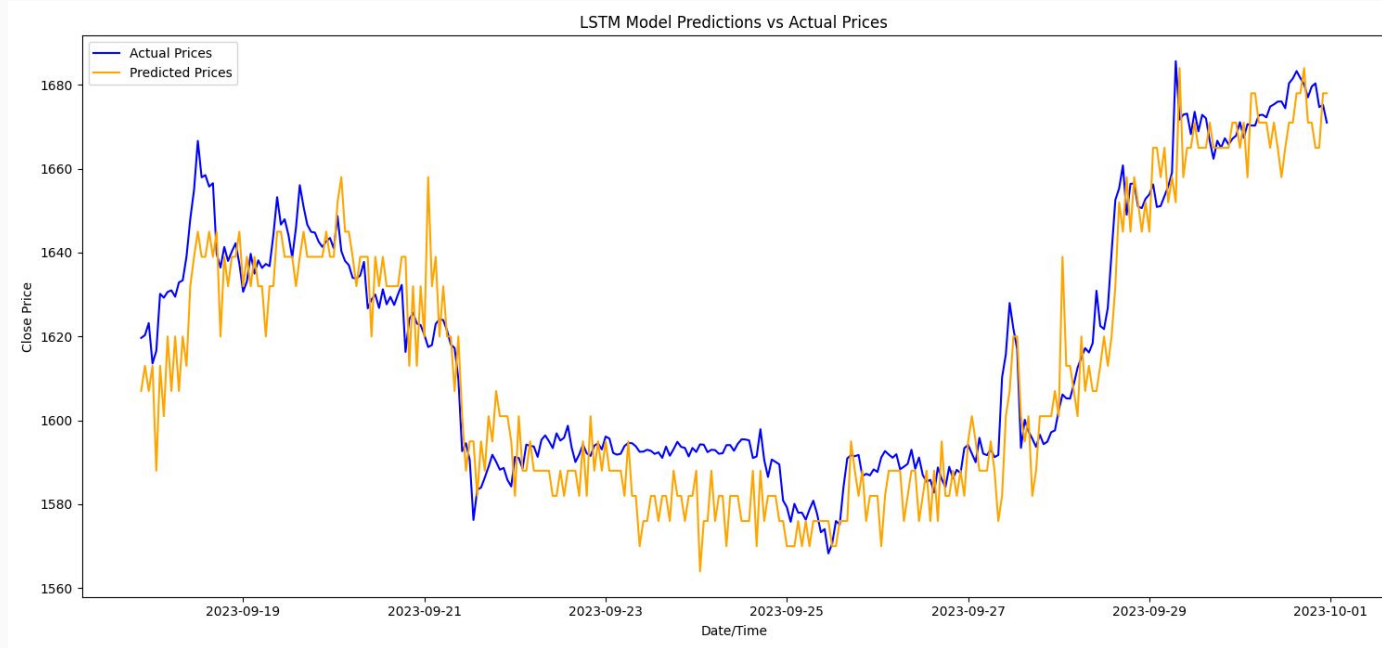
Experimental

		Metrics									
		Run Name	Created	Duration	Models	explained_vari	mae	mape	mse	r2	rmse
		clean-mare-368	7 days ago	10.2min	sklearn	0.89727518...	8.74168253...	0.54038780...	121.069174...	0.87671051...	11.0031438...
		amusing-wren-58	21 days ago	8.0min	sklearn	0.83826531...	10.6369206...	0.65740813...	175.892032...	0.82088225...	13.2624293...
		awesome-bird-301	20 days ago	12.0min	sklearn	0.82506238...	10.9844336...	0.67754872...	175.468597...	0.82460056...	13.2464560...
		dazzling-donkey-423	21 days ago	6.0min	sklearn	0.79188400...	11.8193333...	0.73072497...	216.397936...	0.77963350...	14.7104703...
		tasteful-stork-754	20 days ago	5.8min	sklearn	0.75957379...	14.2881877...	0.87689310...	363.596687...	0.63654661...	19.0682114...
		worried-frog-705	18 days ago	4.8min	sklearn	0.73950870...	14.6729206...	0.90376619...	306.072730...	0.68831414...	17.4949344...
		bright-kite-389	18 days ago	4.8min	sklearn	0.68489873...	15.2413015...	0.94320835...	330.512222...	0.66342645...	18.1799951...
		blushing-cow-278	18 days ago	3.7min	sklearn	0.83418503...	15.8564126...	0.98120456...	350.349238...	0.64322564...	18.7176183...
		amazing-grouse-836	21 days ago	4.4min	sklearn	0.59310808...	16.6504702...	1.03177785...	400.845560...	0.58680728...	20.0211278...
		mysterious-yak-352	20 days ago	8.8min	sklearn	0.88651554...	18.3626213...	1.13215329...	433.583613...	0.56658727...	20.8226706...
		bemused-bug-85	20 days ago	8.1min	sklearn	0.83081262...	18.4685760...	1.13996279...	491.861024...	0.50833283...	22.1779400...
		useful-dolphin-257	18 days ago	4.8min	sklearn	0.66219067...	19.6587619...	1.21450697...	563.494952...	0.42617101...	23.7380486...
		rebellious-pig-500	20 days ago	3.8min	sklearn	0.34485891...	22.3446031...	1.37067080...	996.371016...	-0.0146436...	31.5653451...
		thundering-worm-104	16 days ago	13.7min	sklearn	0.85777219...	22.7182200...	1.40627523...	642.367820...	0.35788536...	25.3449762...
		kindly-mare-445	20 days ago	3.0min	sklearn	0.09337170...	23.5992556...	1.45540016...	911.562189...	0.08879709...	30.1920881...
		masked-swan-24	16 days ago	23.5min	sklearn	0.89180106...	26.1345307...	1.61550066...	784.414940...	0.21589422...	28.0074086...
		merciful-seal-700	16 days ago	7.8min	sklearn	0.58071197...	27.2559047...	1.68658620...	1049.73254...	-0.0689837...	32.3995762...

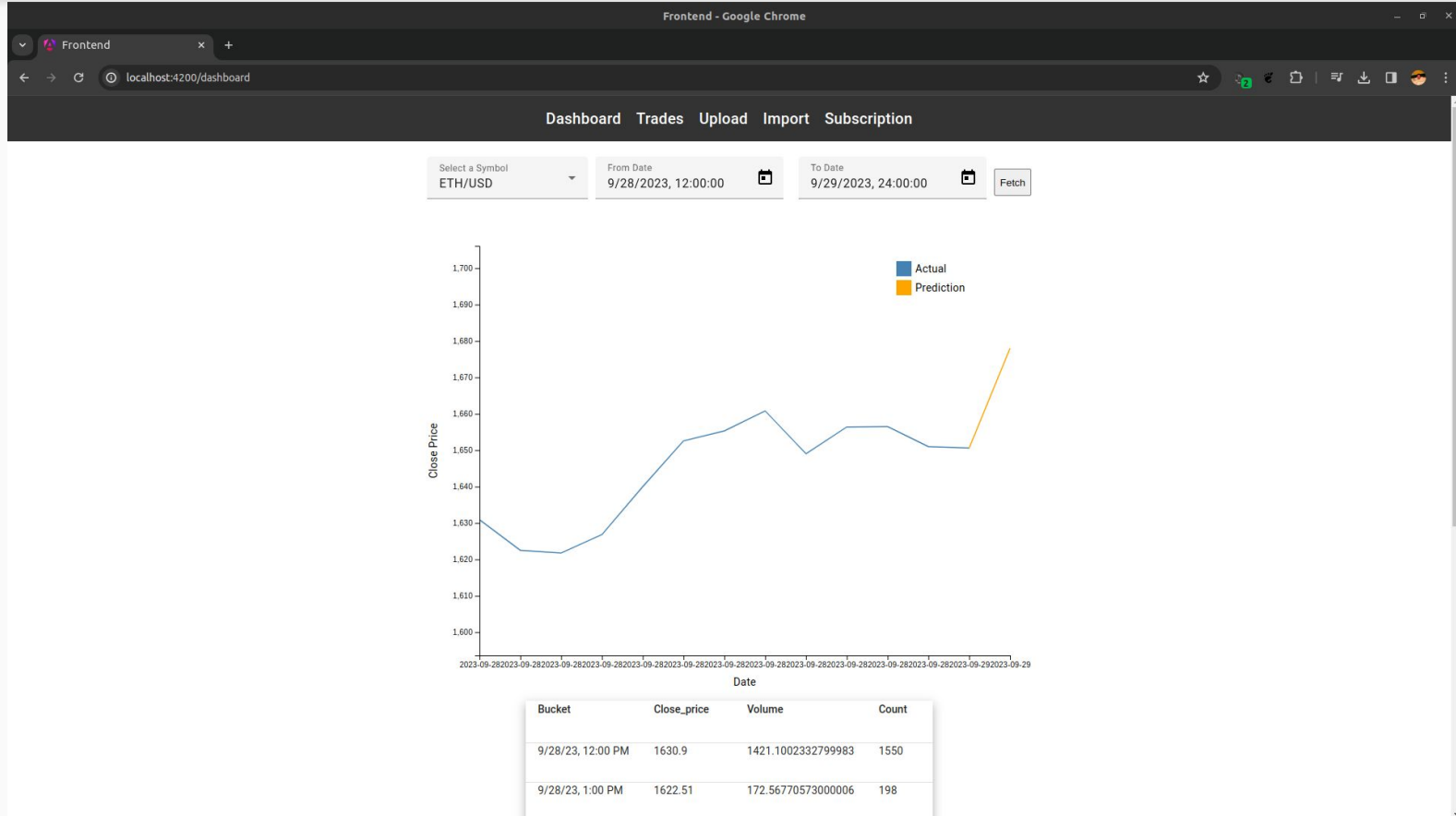
74 matching runs

Model Evaluation: clean-mare-368

Metric	Value
Mean Absolute Error	8.742
Mean Absolute Percentage Error	0.5404
Mean Squared Error	121.07
R2	0.877
Root mean squared error	11.003
Explained Variance	0.897



User Interface: Dashboard



User Interface: Trades

← → ↻ 🌐 localhost:4200/trades

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Dashboard Trades Upload Import Subscription

Select a Symbol
ETH/USD

From Date
1/21/2024, 14:00:00

To Date
1/22/2024, 02:00:00

Fetch

Time	Price	Volume	Side	Order_type	Symbol_id
1/21/24, 2:00 PM	2471.35	0.01993286	b	l	1
1/21/24, 2:00 PM	2471.35	0.00199329	b	l	1
1/21/24, 2:00 PM	2471.35	0.00398657	b	l	1
1/21/24, 2:00 PM	2471.35	0.24228235	b	l	1
1/21/24, 2:00 PM	2471.35	0.03986573	b	l	1
1/21/24, 2:03 PM	2471.35	0.05699959	b	m	1
1/21/24, 2:03 PM	2471.35	0.02502133	b	m	1
1/21/24, 2:04 PM	2471.35	0.00523742	b	l	1

User Interface: Upload & Import

← → ↻ localhost:4200/upload

Dashboard Trades Upload Import Subscription

Upload CSV Files

Upload CSV Files

ETHUSD.csv

Select Files

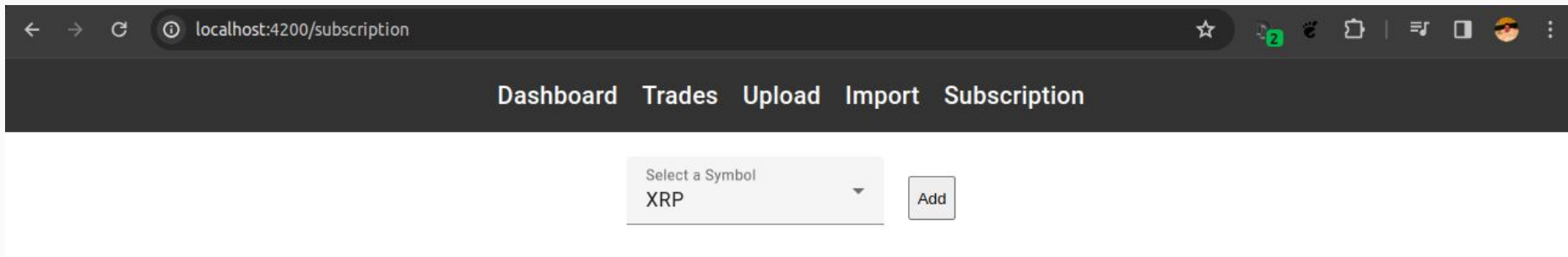
Upload

← → ↻ localhost:4200/import

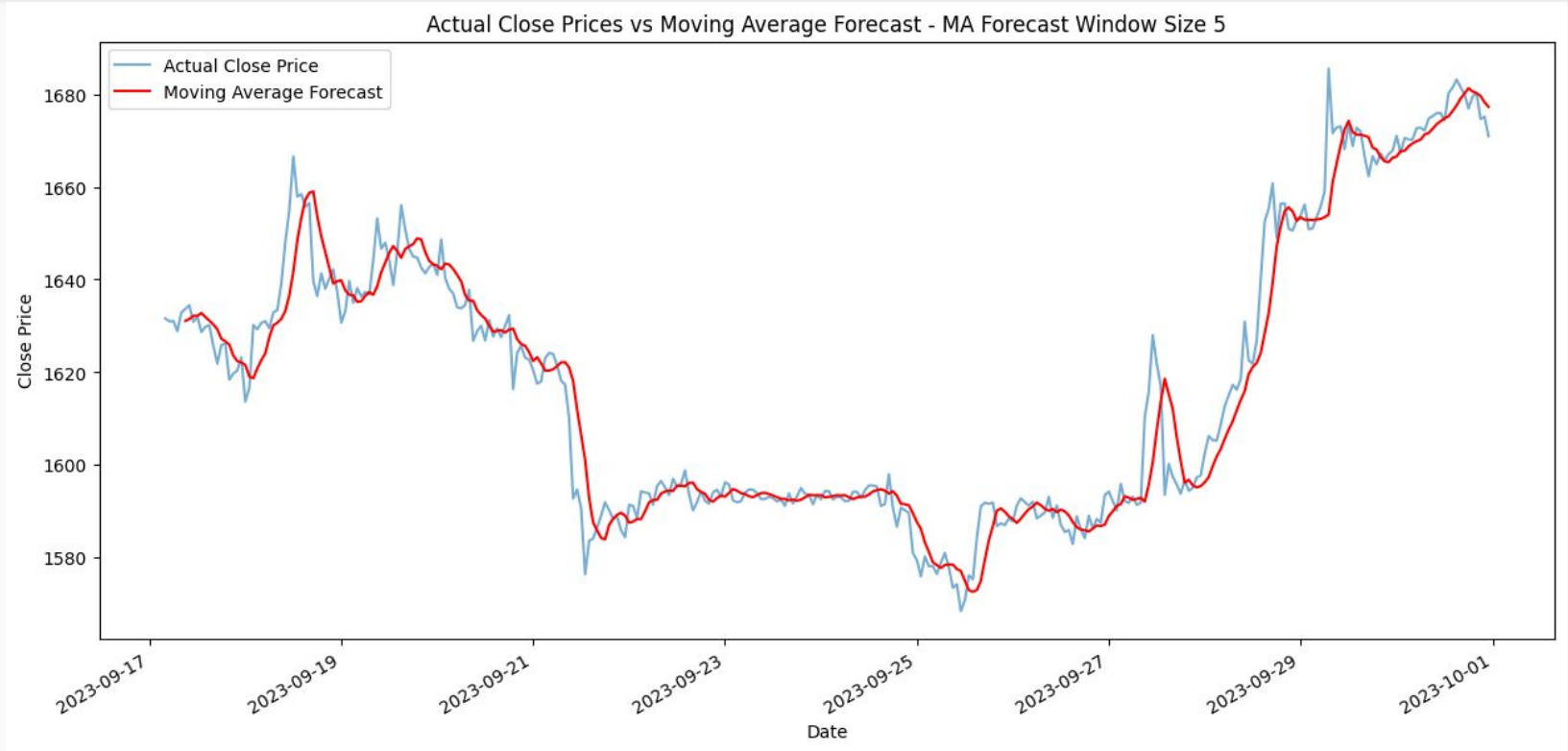
Dashboard Trades Upload Import Subscription

File Name	File Size (bytes)	Uploaded Date	Symbol	Import
ETHUSD_Q1.csv	48947124	1/21/24	Enter Value ETH	Import

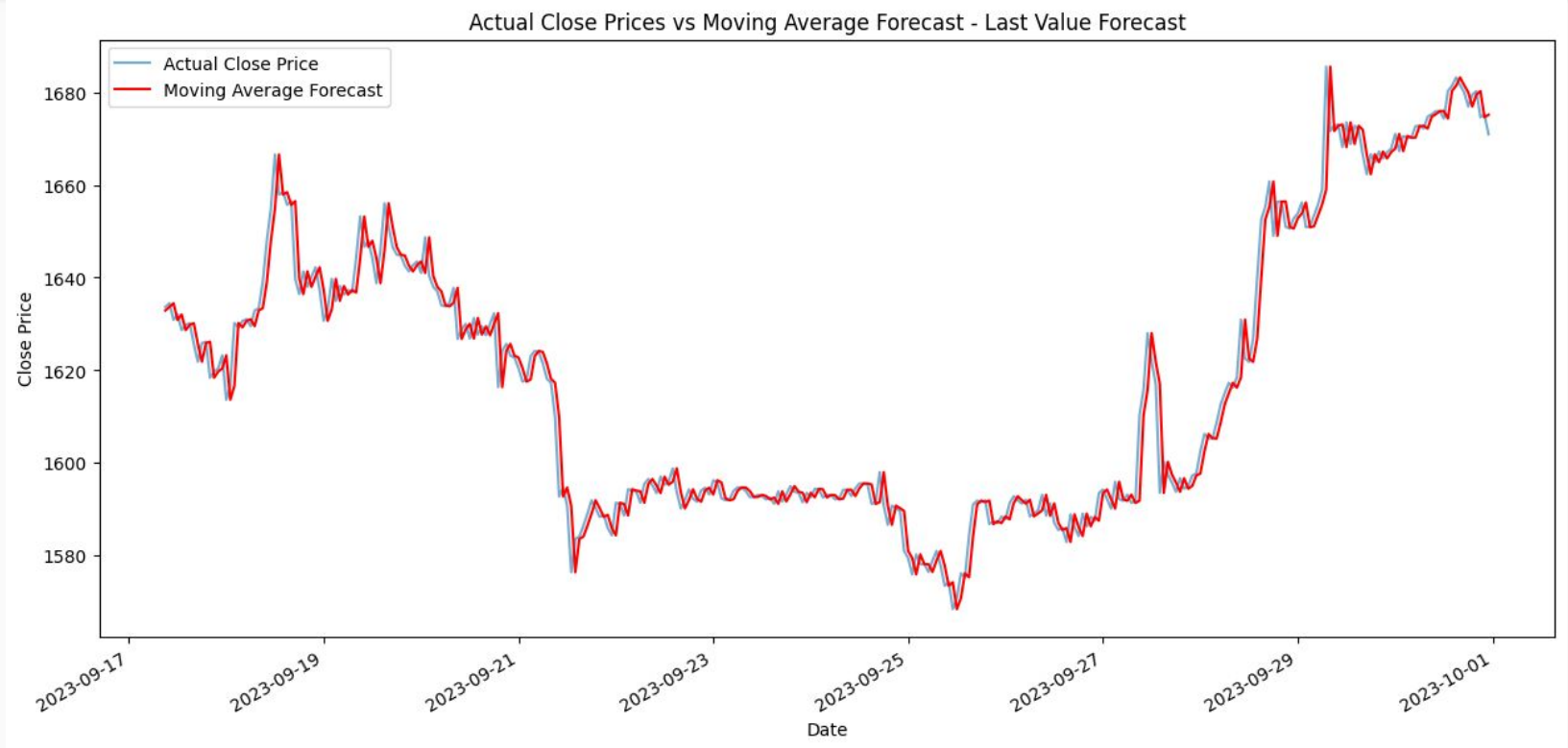
User Interface: Subscription



Evaluation: Moving Average Model



Evaluation: Last Value Model



Evaluation

Metric	Last Value	Moving Average	LSTM (clean-mare-368)
Mean Squared Error	23.52	46.54	121.07
Mean Absolute Error	3.24	4.41	8.74
Root Mean Squared Error	4.85	6.82	11
Mean Absolute Percentage Error	19.94%	27.18%	54.04%
R^2 Score	0.951	0.975	0.88
Explained Variance	0.951	0.975	0.897

Evaluated Timesteps: 327

Improvements:

- Collector Error Handling (Network Outage, etc.)
- LSTM Evaluation additional Metrics: Precision & Recall
- Materialized View - 1min
- Rust and the integration of Python, Tensorflow Bindings

1. Haixu Wu, Jiehui Xu, Jianmin Wang, and Mingsheng Long. Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting, 2022. <https://arxiv.org/abs/2106.13008>